

CRUD v4 — In-Depth Learning Guide

This is the holy bible for React development in this codebase. Written for someone with zero React experience who wants to become an expert fast. Every chapter uses **real code from `react-crud-v4/`** — not made-up examples. Each line is explained: what it does, why it exists, what breaks without it, and what best practice it follows.

How to Use This Guide

If you are brand new to React: Read chapters 1 → 10 in order. Each chapter builds on the previous.

If you are implementing a new feature: Use this as a reference. Jump to the chapter for the file you are working on.

If something is broken: Check chapter 10 (Common Mistakes) first.

Chapters

#	File	What You Will Learn
01	Architecture	How all the pieces fit together, why each layer exists, the data flow from server to screen
02	<code>Services/crud.dtos.ts</code>	Zod schemas, runtime validation, <code>z.infer</code> , why TypeScript alone is not enough
03	<code>Services/crud.types.ts</code>	<code>ItemEditContext</code> , <code>ItemFormSchema</code> , discriminated unions, form validation with Zod
04	<code>Services/crud.services.ts</code>	HTTP calls, <code>ApiClient</code> , <code>zodParse</code> , <code>async/await</code> , Axios, separation of concerns
05	<code>Services/crud.queries.ts</code>	<code>useQuery</code> , <code>useMutation</code> , query keys, <code>staleTime</code> , cache invalidation, <code>enabled</code> , <code>select</code>
06	<code>helpers/</code>	Pure functions, <code>mapItemDtoToRow</code> , <code>buildEditContext</code> , why co-location matters
07	<code>columns/itemColumns.tsx</code>	TanStack Table column defs, <code>buildItemColumns</code> , <code>ActionsConfig</code> , <code>FILTERABLE_COLUMNS</code> , <code>satisfies</code>
08	<code>hooks/useCrudPage.ts</code>	<code>useReducer</code> , <code>useMemo</code> , <code>useCallback</code> , discriminated unions, the orchestrator hook pattern
09	<code>components/</code> + <code>Crud.page.tsx</code> + <code>Crud.module.tsx</code>	Page, Table, Modal, Form, <code>DeleteConfirmModal</code> , <code>StatusPlaceholder</code> , lazy loading
10	—	12 common mistakes with before/after fixes, best practices checklist, full glossary

Quick Reference — React Hooks Used in This Codebase

Hook	Where Used	Purpose
useState	ItemTable, Crud.page	Simple independent local state
useReducer	useCrudPage	Coupled state that changes together atomically
useMemo	useCrudPage, ItemTable	Cache expensive computations, stable references
useCallback	useCrudPage, ItemModal	Stable function references to prevent child re-renders
useEffect	ItemModal	Sync form state when modal opens/context changes
useRef	ItemModal	Store focus target without triggering re-renders
useQuery	crud.queries.ts, ItemForm	Fetch + cache server data
useMutation	crud.queries.ts, ItemModal	POST/PUT/DELETE with success/error callbacks
useQueryClient	crud.queries.ts	Invalidate cache after mutations
useForm	ItemModal	Create form instance with Zod validation
useFormContext	ItemForm	Read form instance from FormProvider context
useReactTable	ItemTable	Create TanStack Table instance
useSearchParams	crud.queries.ts (URL override pattern)	Read URL query params

Quick Reference — File Responsibilities

```
react-crud-v4/
├── Services/
│   ├── crud.dtos.ts      → API shapes only. Zod schemas + inferred types. No UI.
│   ├── crud.services.ts → HTTP calls only. One function per endpoint. No React.
│   └── crud.queries.ts  → React Query hooks. Caching, loading, error,
invalidation.
├── crud.types.ts        → UI types: ItemEditContext, ItemFormSchema,
ItemFormValues.
├── helpers/
│   ├── mapItemDtoToRow.ts → ItemRow type + DTO→Row mapper. Pure function. Co-
located.
│   └── buildEditContext.ts → DTO → edit context. Pure function.
└── columns/
```

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|   └─ itemColumns.tsx → Column defs, DEFAULT_COLUMN_VISIBILITY,
FILTERABLE_COLUMNS.
|   └─ hooks/
|       └─ useCrudPage.ts → All state + handlers. useReducer + useMemo +
useCallback.
|   └─ components/
|       └─ ItemTable.tsx → TanStack Table rendering. Local UI state only.
|       └─ ItemModal.tsx → Dialog + useForm + FormProvider + submit logic.
|       └─ ItemForm.tsx → Form fields. useFormContext + Controller. Self-
fetches data.
|   └─ DeleteConfirmModal.tsx → Confirm dialog. Promise.allSettled for multi-
delete.
|   └─ StatusPlaceholder.tsx → Empty/error/loading states. WCAG aria-live.
└─ Crud.page.tsx → Orchestrator. Calls useCrudPage. Renders children. Zero
logic.
└─ Crud.module.tsx → Lazy-load boundary. Routes. React.lazy + Suspense.

```

The Golden Rules (memorise these)

1. **Each file has one job.** If a file is doing two jobs, split it.
 2. **Components fetch their own data.** Never prop-drill query results down from the page.
 3. **URL param overrides go in `select`.** Never in the page hook or as props.
 4. **Coupled state uses `useReducer`.** Independent state uses `useState`.
 5. **Types are always derived from Zod schemas** via `z.infer`. Never written manually.
 6. **Handlers passed as props use `useCallback`.** Computed values use `useMemo`.
 7. **Multi-delete uses `Promise.allSettled`.** Never `Promise.all`.
 8. **The page component contains zero business logic.** It only calls the hook and renders.
 9. **`ItemRow` is co-located with its mapper.** They always change together.
 10. **Never deviate from this blueprint without explicit approval and documentation.**
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Start reading: [Chapter 1 — Mental Model](#)